

The potential of electric agriculture and mobility for the least-cost rural electrification strategy in Sub-Saharan Africa

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ABSTRACT

Electrification is a crucial driver of rural development in Sub-Saharan Africa (SSA), yet progress is often impeded by low demand density, particularly for grid extension. Despite the promising potential of stand-alone and mini-grid systems, rural areas frequently lag behind due to their low economic viability in electrification. Simultaneously, the mechanization of agriculture and the motorization of transport offer the opportunity to increase economic activity. This publication explores the role of electric vehicles (EVs) and electric tractors (ETs) in rural electrification scenarios.

We present a methodology to quantify the theoretical impact of EVs and ETs on SSA's electrification strategies, applied specifically to Nigeria and Chad. Our simulations in Nigeria reveal that not only supply options with higher capacity are needed, but that EVs have the potential to lower the cost of electricity (LCOE) by an average of 54 %, and ETs have the potential to reduce LCOE by 49 %, depending on local mobility and mechanization demand. Rural areas with significant cropland particularly benefit from using ETs to achieve cheaper electricity access. In Chad, where grid infrastructure is minimal, our assessment shows that EVs and ETs can significantly influence least-cost electrification strategies. By 2030, mini-grids will become the least-cost supply option for 60 % of the newly connected population, compared to just 13 % when only residential demand is considered.

Our methodology advances the widely used ONSSET electrification planning model to estimate the maximum joint potential of electrified agriculture, electric mobility and adapted rural electrification strategies. Our approach involves geospatially estimating energy demands using macroscopic available population, land cover, and infrastructure data. The findings underscore the importance of jointly implementing EVs and ETs and electrification strategies for technology leapfrogging, highlighting the economic viability and potential of EVs and ETs in rural scenarios. The results provide policymakers with quantitative indicators on the impact of the implementation for Nigeria and Chad.

Introduction

Access to affordable and reliable energy is a key enabler for economic growth and development (Ayaburi et al., 2020). Therefore, the United Nations' (UN's) Sustainable Development Goal (SDG) 7 aims to provide by 2030 access to electricity to the 752 million people without any in 2020 (Sub-Saharan Africa (SSA) housing 77% of them) (United Nations Department of Economic and Social Affairs, 2023). Exacerbatingly, UN data shows electrification progress in SSA slowing down due to the increasing difficulty of connecting those hardest to reach in remote rural areas (IEA, 2023). At the same time, electric vehicles are a key technology for mitigating climate change, as CO₂ emissions from vehicles in Africa are currently increasing by around 7% annually (Ayetor et al., 2021), and local air pollution is exacerbated by

the import of mostly used vehicles to Africa (Gautam et al., 2023). Ethiopia recently decided to ban the import of fossil fuel-powered cars in order to accelerate its independence from fossil fuels (Nilsen & Princewill, 2024). With renewable energy in the region abundantly available, EVs enable leapfrogging fossil mobilization. Nevertheless, in addition to the barriers to the adoption of EVs in general as higher one-time cost and limited power supply, electric vehicles in SSA face additional challenges to meet mobility needs, for instance paratransit, demanding road infrastructure and cost of capital. Collett et al. (2021) highlight the need for adapting implementation strategies to the local context. Various authors demonstrate the feasibility of electric mobility in African cities, especially within informal paratransit systems (Ayetor et al., 2023; Giliomee & Booyesen, 2023; Martin et al., 2023). In rural

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settings, where mobility issues are most pressing (Bryceson et al., 2003; Porter, 2002) and fuel supply chains are expensive (Karimu & Dramani, 2021), the literature lacks evidence from EV implementation in funded projects (Allee et al., 2022; Gammon & Sallah, 2021; Siemens Stiftung, 2020) and research (Bugaje et al., 2022; Pawlak et al., 2023; Petit et al., 2017; Rosner & Lienkamp, 2022), while photovoltaics (PV) is available locally. Developing rural areas is particularly crucial for Africa's economic growth, especially concerning food security. Agriculture is a major source of income, contributing up to 40% of gross domestic product (GDP) in the region (Christiansen et al., 2011; Wamalwa et al., 2023). Several studies show that electrification has a positive influence on agricultural output (Amuakwa-Mensah & Surry, 2022; Byaro & Mmbaga, 2022; Omoju et al., 2020). However, with SSA currently experiencing low agricultural productivity (Mrema et al., 2018; Nassirou Ba, 2017), mechanization of crop production (mainly through tractors) is essential to raise productivity in typical smallholder farming (Mrema et al., 2018; Sims & Kienzie, 2016). ETs therefore offer one promising solution to mechanize and raise productivity during fieldwork in accordance to UN's SDG 2, aiming to achieve food security by sustainable agriculture. ETs might not lower barriers for mechanization in general, but have the potential to enable agricultural mechanization and productivity rise in sustainable means: They can address especially smallholder farmer's needs by reducing dependence from fossil fuel (Bagire et al., 2022; Vogt et al., 2021), the electric power train opens up new flexibility in design which allows them to be adapted to the user's context (Götz et al., 2023), they can reduce operational cost (Beligoi et al., 2022; Lagnelöv, Dhillon et al., 2021) and reduce emissions (Lagnelöv, Larsson et al., 2021; Olkkonen et al., 2023). Considering these three critical challenges – electrification, mobilization, and mechanization – both EVs and ETs offer significant synergies with electrification. With the advent of electric vehicles and electric tractors, it is imperative to integrate them into electrification strategies (Dagnachew et al., 2023) to foster sustainable and resilient economic growth.

This publication proposes a top-down perspective on the theoretical implementation of electric vehicles to leapfrog fossil fuel-driven technology, aiming to account for potential synergistic effects between electrified vehicles, machinery, and the optimal electrification approach in rural SSA. Thus we want to provide strategic guidance for policymakers and attract private investment.

State of the art

To reach the goal of quantifying the synergy between EVs and ETs with the optimum electrification strategy, we show the current state of rural electrification in SSA (Section “Rural electrification”), the concept of PUE to make electrification systems economic attractive (Section “Electric mobility & mechanization as productive use of energy PUEs”), and the state of the art on electrification planning tools (Section “Electrification planning”). As our study shows the macroscopic effects of EVs and ETs, Sections “Spatial electric mobility demand estimation” and “Spatial electric agriculture demand estimation” on how to assess the macroscopic demand for those technologies.

Rural electrification

In 2021, 590 million people in SSA still lacked any access to electricity and this number grew over time (Cozzi et al., 2023), even though positive effects of electrification on economic activity have been indicated (Bos et al., 2018). However, such binary classification into access or no access neglects the large effects on feasible activities depending on the multidimensional quality and quantity of supply. Therefore, the World Bank (Bhatia & Angelou, 2015) proposes the MTF for classifying energy access (e.g. including cooking), which is shown in Table 1.

Electrification of under-served areas is, in principle, possible through three distinct modes: creation or extension of centralized grid systems, village-sized minigrids (MGs) and individual standalone (SA) systems (Mandelli et al., 2016). While the former has the highest potential for economies of scale and high power applications but has proven to be slow, both other options are more flexible in deployment (Agutu et al., 2022; Mentis et al., 2016). MGs especially suffer from a multitude of social, technical, economic, environmental, and political challenges in implementation (Babayomi et al., 2023). Economic unsustainability due to high capital expenses and low general consumption is among the most directly inhibiting to unlocking more private investment in the sector (Come Zebra et al., 2021).

Hence, various authors in recent years underline how PUE, for instance through the support of customers' commercial activities with electricity, can enhance the viability of MG and SA solutions—to a limited extent—by increasing energy throughput and generating reliable revenue for the system operator, primarily during daytime hours (Farthing et al., 2023; Lukuyu & Taneja, 2023; Miles et al., 2022; Van Hove et al., 2022). Possible activities are irrigation, agro-processing (e.g. milling, husking, hulling), manufacturing (e.g. tailoring, welding, looming), and services (e.g. gastronomy) (Dagnachew et al., 2023). It must be remarked that seasonal PUEs lead to higher cost of capital due to unused capacity during the off-season (van Hove et al., 2022) as well as Aarakit et al. (2024) states that the social and financial incentives for the operators have to be considered to make PUEs attractive.

Electric mobility & mechanization as productive use of energy PUEs

To our knowledge, only a few authors have proposed, simulated, and/or tested the impact of EVs and ETs on distributed electrification systems as MG and SA. Both EVs and ETs might benefit off-grid electrification modes by generating additional load as PUEs and increasing revenue (Pedersen, 2016). Furthermore, EVs and ETs can suit as a deferrable load when intelligently operated to enable load balancing (Rosner & Lienkamp, 2022).

Electric mobility field trials in rural areas

Gammon and Sallah (2021) pilot this integration using an electric four-wheeler taxi operating from a 4.5 kW_p mini-grid in The Gambia and confirm the symbiosis between MGs and EVs to be beneficial for both the cost of energy access and providing cost-effective transport. The analysis presents the operational cost and expected return for the transport service of the EV and provides preliminary findings for the commercial viability based on surveys. Bugaje et al. (2021) investigate the MG system performance for multiple types of vehicles operated in Kenya by WeTu (Siemens Stiftung, 2020) and highlight the benefit of smart energy management. Allee et al. (2022) report on an electric motorcycle taxi service field trial in a Nigerian village. They find that electric motorcycles operate more cost-efficiently than conventional ones and benefit the MG even more if it is properly sized for the extra load. For certain areas, Lukuyu et al. (2020) propose electric fishing boats as a productive use for MG and SA systems by identifying the user needs and sizing the MG according to the boat operations, contributing to the sustainable development of coastal communities and unlocking economic benefits.

Electric agricultural mechanization field trials

Bagire et al. (2022) propose ETs to be 25% cheaper to operate in SSA, but do not shed any light on energy supply. The authors propose the ET as an efficient source of farm power to raise the availability of agricultural machinery in SSA. The tractor model tested was found to be feasible, adding environmental value and improving agricultural productivity. However, there is no investigation on how the tractor can be supplied by electricity in rural African circumstances. Hence, our research group (Götz et al., 2023) proposes to look into a holistic concept of electricity supply and ETs in order to exploit the potential

Table 1
Household electricity access classification in the MTF (Bhatia & Angelou, 2015).

Access level	Indicative appliances	Yearly consumption
Tier 1	Task lighting, phone charging & radio	8 kWh/capita
Tier 2	General lighting, air circulation & television	44 kWh/capita
Tier 3	Tier 2 & light appliances (e.g. washing machine)	160 kWh/capita
Tier 4	Tier 3 & medium appliances (e.g. water pumping)	423 kWh/capita
Tier 5	Tier 4 & heavy appliances (e.g. air conditioning)	598 kWh/capita

of agricultural electrification. The study sees the need to include the energy supply side in the design of new ETs to ensure feasibility, as they can, at worst, be a heavy burden on local power grids or, at best, act as a support for local grids when used as a buffer.

While not covering ETs, Korkovelos et al. (2022) do see the need for accurately estimating electricity requirements for irrigation as a PUE in SSA. Their “agrodem” open source model quantifies the electricity needs for pumping water by using geospatial data (e.g. cropland, climate, and soil data). The Open Source Spatial Electrification Tool (OnSSET) is used to derive the changes in the electrification strategy with the resulting energy demand. The paper underscores the importance of integrating energy needs in agricultural planning.

Shirley et al. (2021) explore the spatial linkage between energy access and opportunities to electrify primary processing and post-harvest activities. The authors show, with a case study in Uganda on maize and coffee, those areas that offer the most considerable potential. Wamalwa et al. (2023) analyze the joint deployment of MGs and smallholder irrigation for economic potential: By applying their model to three rural case studies, the results show that irrigation loads can lower LCOE of MGs by up to 47%, depending on the quantity and distribution of irrigable land. The results highlight the benefit of co-planning of MGs and irrigation projects and can be used for high-level decision-making.

All previously-shown simulation and field trial studies highlight the impact that electrification of mobility and agricultural machinery has on the least-cost electrification strategy, but also the difference it can make to the economic feasibility of MG and SA systems. Together with Dagnachew et al. (2023), we want to show with our investigation the importance of the consideration of such increased demand already in planning electricity systems. Furthermore, Lukuyu and Taneja (2023) highlight that strategies and measures to stimulate electricity demand are becoming increasingly essential to ensure socio-economic development in the entire SSA region and emphasize the need for further investigation of demand stimulation strategies to enable sustainable growth.

Electrification planning

To allow for widespread adoption of such productive use, distributed electrification strategies must consider them in electrification planning, both locally and on aggregated levels to direct funds and efforts to the most useful area and identify the most promising innovations (Dagnachew et al., 2023). Geospatial information system (GIS) tools simplify considerations in the temporal dimensions to be able to determine the least-cost pathways for many settlements on a large scale with limited information available (Szabó et al., 2011). Szabó et al. (2011) first present such an investigation for the African continent. Multiple following studies expand both the geographic scope and technical depth, forming them into frameworks adaptable to new problems or countries (Bertheau et al., 2017; Bissiri et al., 2020; Ciller & Lumberras, 2020; Egli et al., 2023; Isihak, 2023; Korkovelos et al., 2019; Mentis et al., 2017; Müller et al., 2016; Szabó et al., 2016). Most of these studies employ one of the three such frameworks, namely the Open Source Spatial Electrification Tool (OnSSET), the Reference Electrification Model (REM), and Network Planner (Bhattacharyya & Palit, 2021).

Among those, “OnSSET has become an ubiquitous geospatial electrification modeling tool for academia and international organizations

working in electricity access in developing countries, such as the Global Electrification Platform (GEP) by World Bank’s Energy Sector Management Assistance Program (ESMAP)” (Egli et al., 2023), which is why we choose it for the upcoming investigations.

Spatial electric mobility demand estimation

The previous section showed that the electrification of rural areas and PUEs have a mutual influence on each other. Hence, spatial electrification planning requires a spatial assessment of the demand for EVs and ETs.

In literature, five main methods exist for spatial allocation of EV charging demand, most often coupled with placement algorithms for creating optimal charging networks. These are in order of increasing complexity (a) analytical approaches spatially dividing total charging demand *top-down* based on empirical relations (Adenaw & Lienkamp, 2020; Frade et al., 2011), (b) flow based methods placing charging stations near maximum traffic flows (Lim & Kubry, 2010), (c) regression methods from structural data such as Points of Interest (POIs) (Straka et al., 2020), (d) data based approaches using destinations from recorded vehicle/personal movement traces (Andrenacci et al., 2016), and (e) complex simulations encompassing mobility demand, vehicle, and behavioral data (Shepero & Munkhammar, 2018; Viswanathan et al., 2016).

Concerning large-scale GIS approaches, Prakobkaew and Sirisumrannukul (Prakobkaew & Sirisumrannukul, 2022) propose discretizing the study area into smaller cells to allocate estimated numbers of EVs to them using registration data, make assumptions on daily driven distances and consumption per vehicle class (Li et al., 2020; Shepero & Munkhammar, 2018; Vigano et al., 2021) and place chargers based on the charging demand estimated in this manner.

However, mobility data is extremely scarce in SSA, highlighting the need for more public available data in the region (Collett & Hirmer, 2021). As also vehicle registration data is not publicly available, and mobility behavior is vastly different from other regions with high paratransit penetration (Collett et al., 2021; Evans et al., 2018), and the large target area is negating complex assumptions and models, we place our demand estimations on literature and experiences to the best of our knowledge accompanying uncertainties. We favor a simple analytical method with constant consumption values for this publication that can be refined when additional data becomes available.

Spatial electric agriculture demand estimation

Unlike EVs, ETs are still a relatively new technology and have only been deployed in SSA in pilot runs (GIZ, 2023). To our knowledge, no published research focuses on spatially estimating either the regional demand in tractors generally or the charging demand for broad adoption of ETs.

Agricultural processes are identified as energy-intensive. As mentioned in Section “Electric mobility & mechanization as productive use of energy PUEs”, the “agrodem” model from Korkovelos et al. (2022) estimates the energy needed for irrigation based on crop extent, the appearance of the surface, groundwater and climate data in combination with the required irrigation target. Farthing et al. (2023) spatially estimate the load for various agri-processing steps in 100 km grid cells per crop extent or average animal appearance. The included MG case

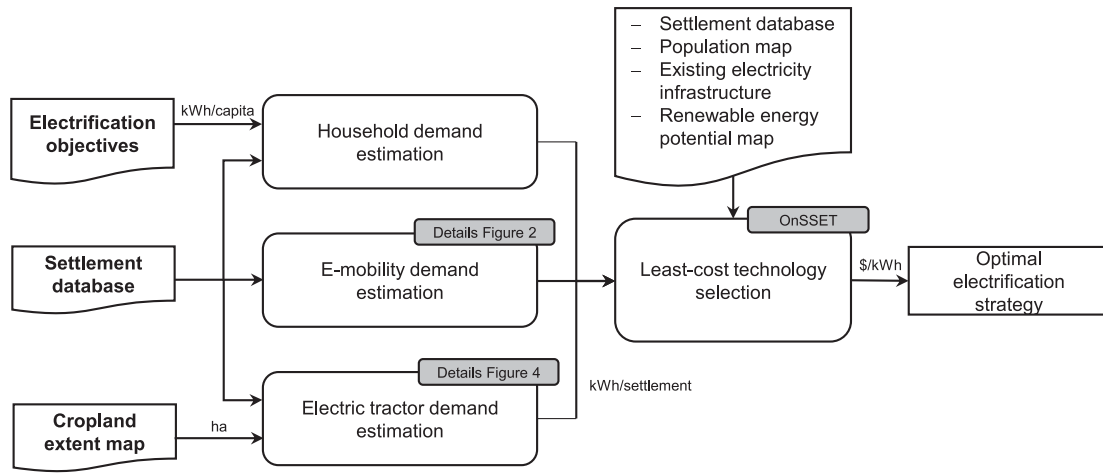


Fig. 1. Methodology to derive optimal rural electrification strategy with the PUEs of electric vehicles and electric tractors.

studies demonstrate that the impact of these PUEs is highly dependent on the load profiles and their seasonality. In the case of Farthing et al. (2023) the LCOE was not significantly worsened, but that it remains a challenge how location-specific market potential of agricultural PUEs can complement distributed power systems. Wamalwa et al. (2023) contrarily find that joint deployment of MGs with smallholder irrigation can lower the unit cost of MG energy by up to 47%. The energy demand for irrigation is estimated on the extent of cropland in the targeted area. In summary, understanding the potential agricultural energy load in SSA is essential if agricultural productivity is to be stimulated by electrification and symbiotic potential is to be unlocked with electrification strategies.

Research gap and contribution

To the best of our knowledge, few authors show the interaction between off-grid electrification and electric mobility PUEs on a local level as well as electric agricultural machinery in general.

None of them consider the sizeable impact electric mobility and agriculture can have on the large-scale electrification strategy through the induced shift in electricity demand. As this information is crucial for the widespread future adoption of EVs and ETs for sustainable mechanization and transportation and the fulfillment of SDG 7, we show the following contribution to the state of the art:

- (i) Geospatial demand estimation of mobile PUEs such as ETs and EVs in electrification strategy assessment with limited available input data
- (ii) Derivation of the macroscopic impact of ETs and EVs on the electrification strategy on a country level

Methodology

We aim to show the interaction between rural EV and ET application as PUEs and the least-cost electrification strategy on the example of Nigeria and Chad. Fig. 1 shows the overall procedure to derive the optimal electrification strategy.

Our core methodology is the spatial estimation of maximum electricity demands for EVs and ETs (Sections “Spatial electric mobility demand estimation” and “Spatial electric agriculture demand estimation”). The residential electricity demand is considered independently from those PUEs with target MTF tiers of 4 for urban and 1 for rural households (Table 1, abbreviated as “U4R1”). By incorporating the combined electricity demand for all, we advance the Open Source Spatial Electrification Tool (OnSSET) by adding the input of EVs and ETs demand, being explained in Section “The Open Source Spatial Electrification Tool”.

Spatial electric mobility demand estimation

We focus on personal mobility in rural areas, for which (a) motorcycle paratransit is the primary mode of motorized transport (Pawlak et al., 2023), (b) travel demand can be estimated more accurately per household and (c) charging infrastructure and demand can be assumed to be located close to customers’ homes.

Given spatial mobility demand data scarcity, we set up our estimation model analytically and pattern-based, dependent on few variables such as the population distribution and the distance to the next major road from Khavari et al. (2021), and parametrize it with highly localized case studies (Bryceson et al., 2008; Calvo, 1994; Republic of Ghana, 2013).

It hits the right order of magnitude in demand, which we deem sufficient for judging the magnitude of the continent-wide effects, but incorporates uncertainties which can be just refined by the availability when additional data becomes available. Fig. 2 visualizes the process to derive the charging demand per settlement based on the assumed mobility pattern (Table 2).

To incorporate the spatial influence in the mobility demand, the weekly roundtrip mobility demand per household $d_{hh,week}$ is composed of a spatial and a constant share (Eq. (1)). The former covers trips to and from the closest paved road (index $road$), as those junctions represent important hubs for further trips to markets and/or reach further destinations using means of transportation available from there. Allee et al. (2022) find the closest road to be the most frequent destination in their electric motorcycle field trial, highlighting the importance of this factor visualized in Fig. 2(a) for the case of Nigeria. We limit the distance considered to 50 km to realistically model distances covered in a reasonable time. The latter constant demand component covers trips to workplaces (formal or informal), primary schools, health facilities, and markets (Set D). It is parameterized as the mean distance from collected studies (Bryceson et al., 2008; Republic of Ghana, 2013) and their frequency (Calvo, 1994) summarized in Table 2. Especially, to account for local circumstances, contextual parameters and reduction of uncertainties in quantity, our study allows for future input of updated and more detailed data.

$$d_{hh,week} = 2 \left(\underbrace{f_{road} \cdot d_{road}}_{\text{spatial}} + \underbrace{\sum_{i \in D} f_i \cdot d_{i_c}}_{\text{constant}} \right) \quad (1)$$

Eq. (2) transforms mobility demand to energy demand per settlement as in the right half of Fig. 2. To get an estimation of the maximum strategic potential, we assume an EV penetration rate or modal share ω in personal transport of 100% and an occupancy n_{occ} of one passenger per round trip. The electric demand for such a mobility pattern further

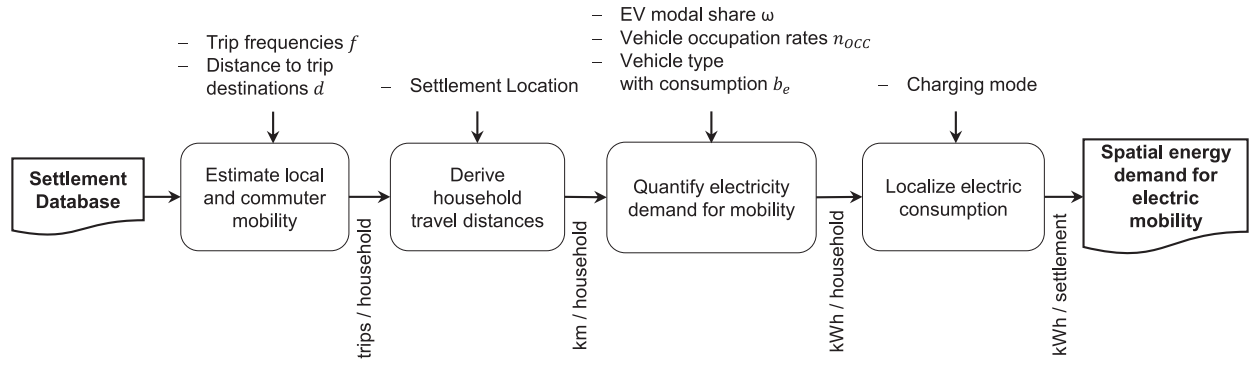


Fig. 2. Spatial estimation of electricity demand for rural electric mobility.

Table 2

Estimated trip distances and frequencies per household.

Destination	Distance	Frequency f_i
Work	4.4 km	6x/week
School	1.6 km	6x/week
Health	5.9 km	1x/week
Market	4.1 km	2x/week
Road	d_{road}	2x/week

depends on the vehicle type's consumption b_e . In accordance with other studies (Section "Spatial electric mobility demand estimation"), we assume a constant value of 48 Wh/km (Bugaje et al., 2021) for electric motorcycles. The resulting yearly (factor $n_{weeks} = 52$) charging demand per settlement E_{sett} is calculated through the estimated number of households n_{hh} and usable for the calculations of OnSSET.

$$E_{sett,mob} = \omega \cdot n_{weeks} \cdot d_{hh,week} \cdot \frac{b_e}{n_{occ}} \cdot \underbrace{\frac{n_{p,sett}}{n_{p,hh}}}_{=n_{hh}} \quad (2)$$

Spatial electric agriculture demand estimation

Utilizing Copernicus Land Cover data (Buchhorn et al., 2020), which classifies "land covered with temporary crops followed by harvest and bare soil periods", we spatially locate cropland suitable for ET deployment in a 100×100 m raster format. While energy demand for mechanized agriculture is highly influenced by various factors (e.g. crop type, soil type, crop yields, and harvest calendar), the demand for ETs primarily depends on the area cultivated with them. Given the dominance of smallholder farming systems in SSA that are characterized by irregular field patterns, small field sizes and mixed cropping (Ibrahim et al., 2021), we assume that all cropland is a potential area where the tractors could be deployed. This ensures that we account for the maximum impact widespread employment of ETs can have on the electrification strategy. Fig. 5(b) illustrates the distribution of cropland across Nigeria, with a notable concentration in the northern regions of the country.

As second data set (see Fig. 3) we use is the settlement database from Khavari et al. (2021) to derive the spatial energy demand for ETs. In focusing on smallholder farming prevalent throughout SSA, the cropland is primarily cultivated by inhabitants residing in the nearest settlement clusters. Our method allows for the connection between the cropland and the nearby cropland areas: We spatially allocate the cropland to the closest settlement by employing Voronoi polygons generated from settlement centroids as shown in Fig. 4. These polygons, created using Quantum GIS (QGIS) 3.32.0, divide the area by calculating the Euclidean distance between the settlements' centers to all neighboring centers and split the distance equally. The Voronoi polygons and the Copernicus Land Cover data are superimposed, hence, each cropland

is allocated to the closest settlement. Cropland cells intersecting by polygon borders are split according to areal size.

By allocating cropland to settlements, we ensure that it is worked from the nearest smallholder farms. This implies that each tractor is charged using the infrastructure located at the smallholder farms, which are allocated to the closest settlements. This approach maximizes the benefits from localized charging infrastructure, strategically positioned to support efficient energy use and enhance productivity across smallholder farms.

As the consumption of the ET depends on its operation, we derive that demand through an average ET areal consumption b_e of 33.2 kWh/ha from Lajunen (2022). The charging demand considers that depending on the crop type and the cultivation, the tractor can be used for multiple yearly operations n_{op} per field. For this study, we assume four operations (plowing, cultivating, seeding, and harvesting) in one yearly crop cycle as an average. The method allows for including more detailed data in a future step to account for actual tractor operations. As with electric transportation in Section "Spatial electric mobility demand estimation", we consider the maximum need for ETs when the entire cropland is covered by ET operation. Hence, this study's calculated with the modal share ω amounting to 100%. With the resulting yearly demand we advance the input in OnSSET per each settlement.

$$E_{sett,agri} = A_{crop,sett} \cdot n_{op} \cdot b_e \cdot \omega \quad (3)$$

The Open Source Spatial Electrification Tool (OnSSET)

Our investigation advances OnSSET, as it misses the PUE for EVs and ETs and neglects their impact and potential for the least-cost electrification strategy in SSA. OnSSET calculates the least-cost (i.e. least LCOE) electrification mode (among grid extension, hydro/PV/wind/hybrid MG, and PV-based SA systems) for each non-electrified settlement for household electrification (defined as connected $100 \text{ m} \times 100 \text{ m}$ raster cells with nonzero population and no night lights Khavari et al., 2021). In accordance with SDG 7, this study set the goal of 100% household electrification up to the desired household demand of U4R1 (see Table 1) and the PUEs of EVs and ETs until the target year 2030.

The decisive parameter for the selection of technology in OnSSET is the LCOE: Eq. (4) calculates it by adding and discounting (at a rate r) investment (I), operational (O), and maintenance (M) cost (C) at every timestep (t) and setting it in relation to the discounted energy (E) sold to generate a measure of economic efficiency (Come Zebra et al., 2021).

$$LCOE = \frac{\sum_{t=0}^m \frac{C_{I,t} + C_{O,t} + C_{M,t}}{(1+r)^t}}{\sum_{t=0}^m \frac{E_t}{(1+r)^t}} + \underbrace{LCOE_{grid}}_{\text{grid extension only}} \quad (4)$$

Investment cost for off-grid technologies is estimated using fixed specific costs and required system capacity $P_{I,off-grid}$, which in turn is

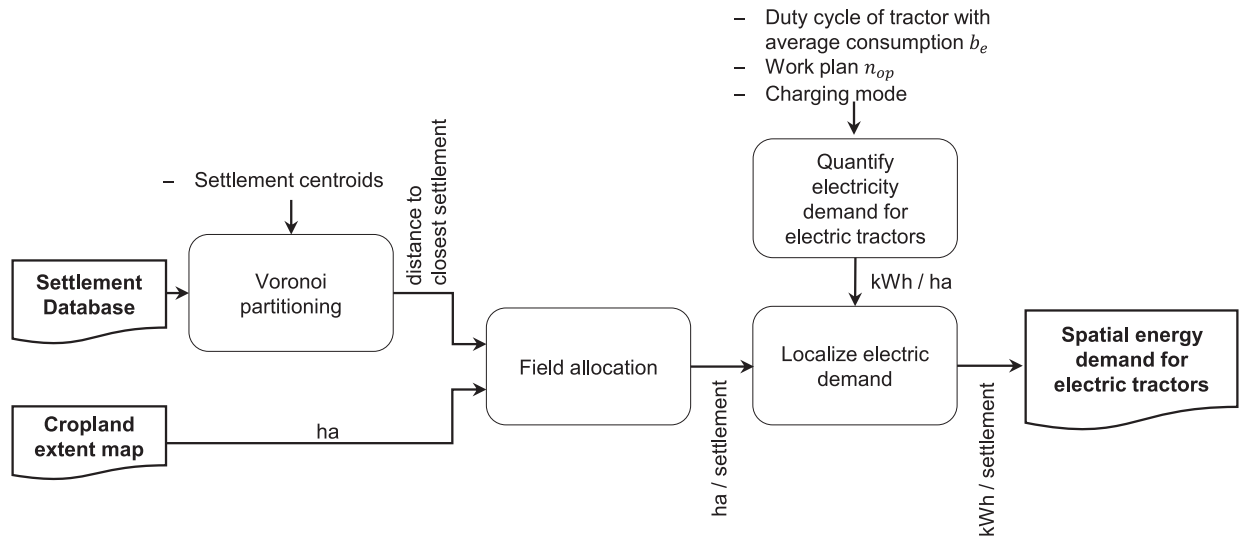


Fig. 3. Spatial estimation of electricity demand for electric tractors.

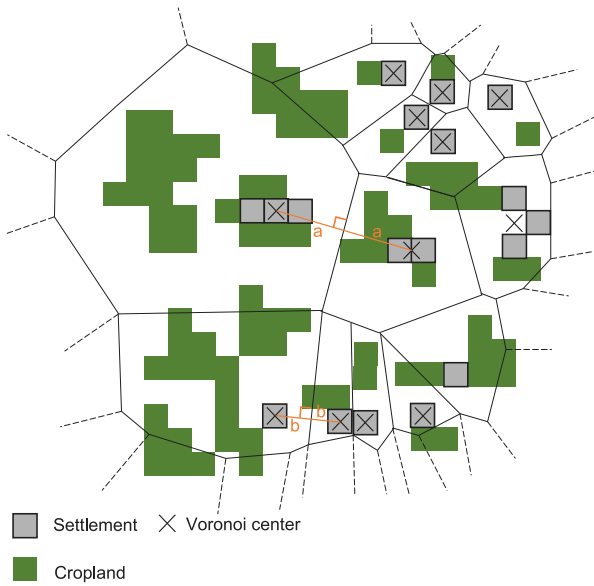


Fig. 4. Allocation of cropland to nearest settlement by Voronoi cells.

determined by average yearly ($t = 8760$ h) demand P_{avg} divided by the base-to-peak factor to get the peak demand P_{peak} , and the technology's capacity factor k_C as per Eq. (5). Average demand estimates vary based on electrification tiers in Table 1. While LCOE for off-grid options can be calculated independently, grid expansion follows an iterative algorithm as transmission cost lowers if settlements around are already being electrified through the grid. Settlements closer than 50 km tend to get forced to connect to the existing grid. The LCOE of the national grid is treated as an input in Eq. (4), whereas the enhancement of power plant capacity and the costs for the transmission lines to connect settlements and their distribution networks are treated as investment costs (Mentis et al., 2017). It should be noted that current expansion plans, as opposed to the existing grid infrastructure, have not been considered in this analysis, as it focuses on identifying the most cost-effective options, which may not necessarily reflect ongoing or planned grid expansion efforts. Furthermore, we set global techno-economic parameters for the analysis according to GEP (World Bank Group, 2021), such as the expected electricity generating cost from the central grid individually per country (Pappis, 2016; Taliotis et al., 2016; World Bank Group, 0000), the capital cost of photovoltaic systems, and the

diesel price (Szabó et al., 2011). None of the settlements is forced to be connected to the grid. All parameters can be found in "Appendix".

$$P_{I,off-grid} = \frac{E_{sett}}{\eta_{dist} \cdot t} \cdot \overbrace{k_{B2P}^{-1} \cdot k_C^{-1}}^{=P_{peak}} \cdot \underbrace{k_C^{-1}}_{=P_{avg}} \quad (5)$$

Results

We have chosen to examine the detailed results for Nigeria and Chad in Section "Optimum technology split", as electrification is of great importance for these countries and is very different in comparison: The former is the most populous country in Africa and exhibits a rural electrification rate of 26%. Even though the grid is reaching far across the country already, 86 million people are left to be electrified, making it the African country with the highest non-electrified population (World Bank Open Data, 2021). Nigeria has 29 million hectares of cultivated area, with the main temporary crops such as maize, cassava, guinea corn, yam, beans, millet, and rice having a high potential for the use of tractors (Buchhorn et al., 2020; FAO, 2024). In contrast, Chad, with its minor grid infrastructure and current rural electrification rate of 1.3% is heavily dependent on the off-grid electrification modes, with PUEs playing a particularly important role in making them profitable (World Bank Open Data, 2024b). In the following visualizations, each Voronoi cell is entirely colored for better discernibility of the chosen electrification mode of a certain location and does not convey any information about its viability (i.e. LCOE) or population size.

Optimum technology split

Nigeria

Fig. 6(a) shows the least-cost electrification strategy among the covered modes (Section "The Open Source Spatial Electrification Tool") for Nigeria in 2030 considering only household electricity demand as per Section "Methodology". It is evident that only highly populated urban areas should be and mostly are already connected to the grid, whereas SA systems cover the near totality of the country's area. MGs do not play a noticeable role in the strategy, as low demand in MTF tier 1 prevents them from being economically viable. This makes investments in PUE supporting infrastructure seem unattractive, leading to low production and economic activity perspectives.

Including the demand for electric mobility in the electrification strategy as per Section "Spatial electric mobility demand estimation"

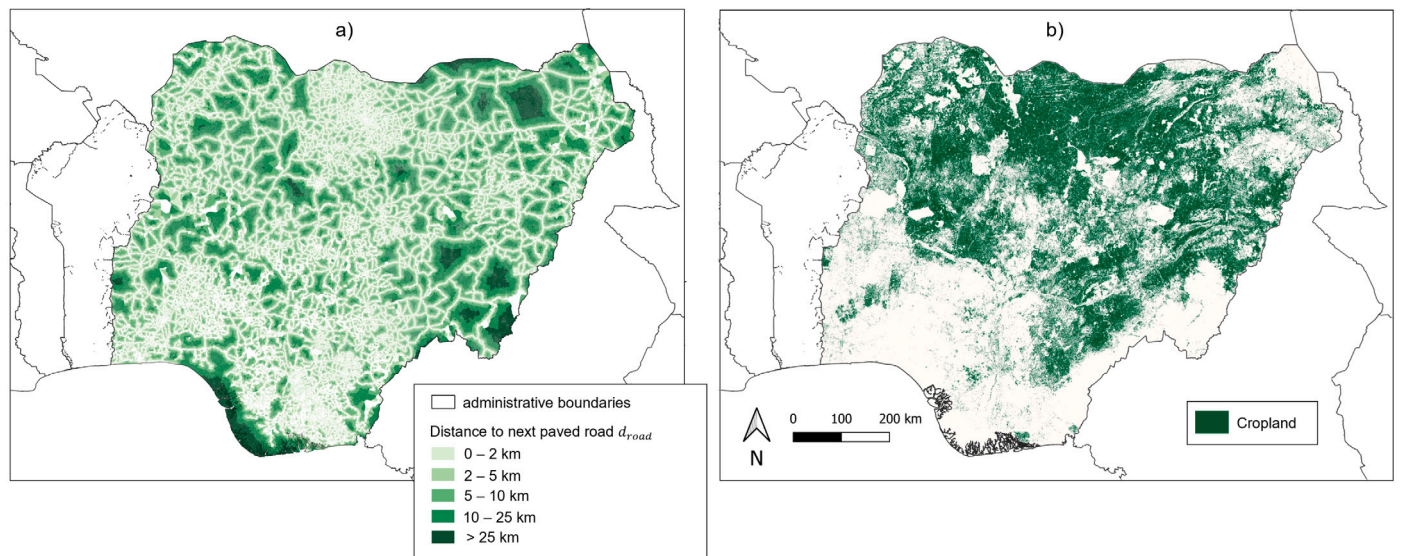


Fig. 5. (a) Distance to the closest paved road per settlement in Nigeria. (b) Cropland in Nigeria, according to Copernicus Global Land Cover.

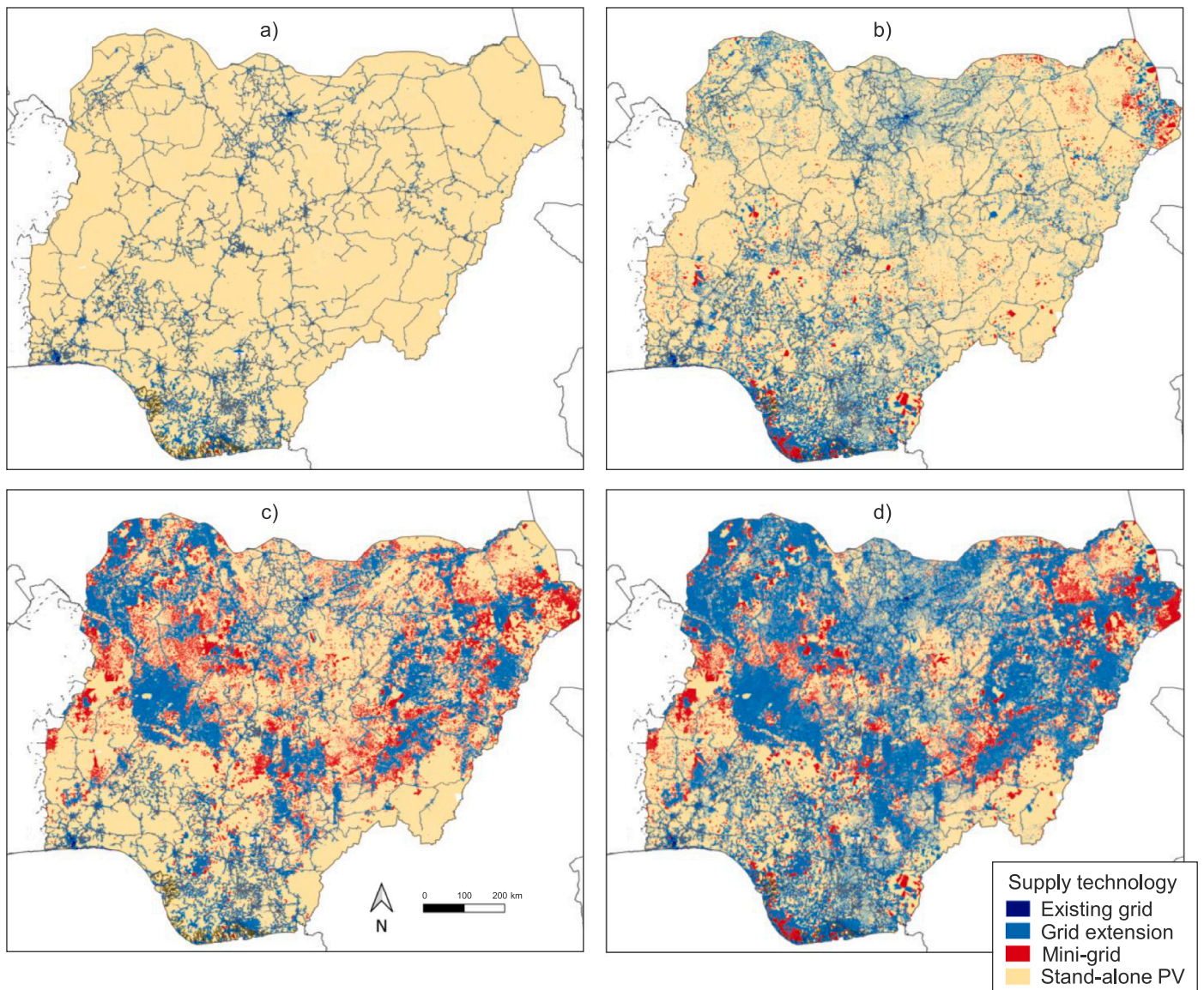


Fig. 6. Optimum electrification strategies for Nigeria until 2030: (a) Household demand only (U4R1) (b) EVs & U4R1 (c) ETs & U4R1 (d) all combined.

in Fig. 6(b), the supply option changes in areas where mobility demand is estimated to be high: Electric mobility changes the supply mode for 56% of the unelectrified population, highlighting the importance of spatial PUE consideration for a sustainable electrification strategy and to attract external infrastructure investment. 53% of the total mobility demand in the yet unelectrified areas is covered by new grid connections, about 3% by MGs and 28% by SA systems. The supply option changes where either settlement population and/or distances to the closest road are high (e.g. on the southern coast). This is because the high investments associated with the grid and MGs are offset by the economies of scale resulting from the more significant amounts of energy sold, which leads to a reduction in LCOE. Therefore, settlements experiencing high relative demand increase change their least-cost supply mode.

On average, settlements covered by grid extension have a higher absolute demand and are closer to the existing grid. In contrast, MGs excel at providing lower LCOE in more remote areas that are still densely populated.

Fig. 6(c) shows the effect of ETs as per Section “Spatial electric agriculture demand estimation” in addition to the baseline household demand. Overall, the least-cost electrification mode changes for 15% of the population but 56% of cultivated area compared to the baseline. This also leads to 56% of the total ET energy needs to be covered by grid extension or MGs, and the remainder by SA PV systems.

ETs significantly impact rural areas regardless of population density, as only available cropland directly influences demand, favoring hybrid MGs and grid extension as the most cost-effective supply modes in agriculturally dominated areas such as northern Nigeria. This supports the hypothesis that PUEs require supportive and scalable electrification strategies and offer added value in the electrification of more remote areas with lower population density. As cropland tends to be further away from the existing grid but is assigned high electricity demand, the supply option of MG gains major importance. As agriculture is the dominant economic activity in rural SSA, ETs’ potential makes it more economical to electrify households further away from current infrastructure and connect those hardest to reach.

The combination of both PUEs in Fig. 6(d) shows the widespread electrification by grid extension, as the higher demand does not just favor the grid extension but also leads to connecting nearby settlements to the newly installed grid. Still, MGs play a significant role of 3% of the newly connected population in the electrification strategy.

Chad

Fig. 7 shows the electrification in the case of the country Chad until 2030. Compared to the population density of 234 people per square kilometer of Nigeria in 2021, Chad only has a value of 14 people per square kilometer (World Bank Open Data, 2024c). Due to the limited existing grid infrastructure and sparse population, the most cost-effective electrification strategy for household demand is almost exclusively provided by stand-alone PV systems (84%). In terms of electric mobility, the settlements in the north of Chad, which are far from the paved road network, have a particularly high per capita demand. The remote environment favors using MGs, which need to supply more than half of the new connected population (51%). In the case of the PUE of electric tractors, almost the entire cropland of 3.9 million hectares, mainly located in the southern part of the country, is most economically supplied by MGs (Fig. 7(c)) (Food and Agriculture Organization of the United Nations, 2024). Even with the combination of EVs and ETs (Fig. 7(d)), none of the PUEs has such a high demand that it is economically reasonable to implement a central grid infrastructure.

The electrification strategies up to 2030 are highly dependent on the unique spatial conditions of each country, leading to varied approaches for the two exemplary nations. Fig. 8 shows the case of Nigeria, where 101 million people are affected, not having access to electricity in 2020. As Nigeria already has an existing grid infrastructure, the PUEs of electric mobility favors grid extension close to the existing grid. In

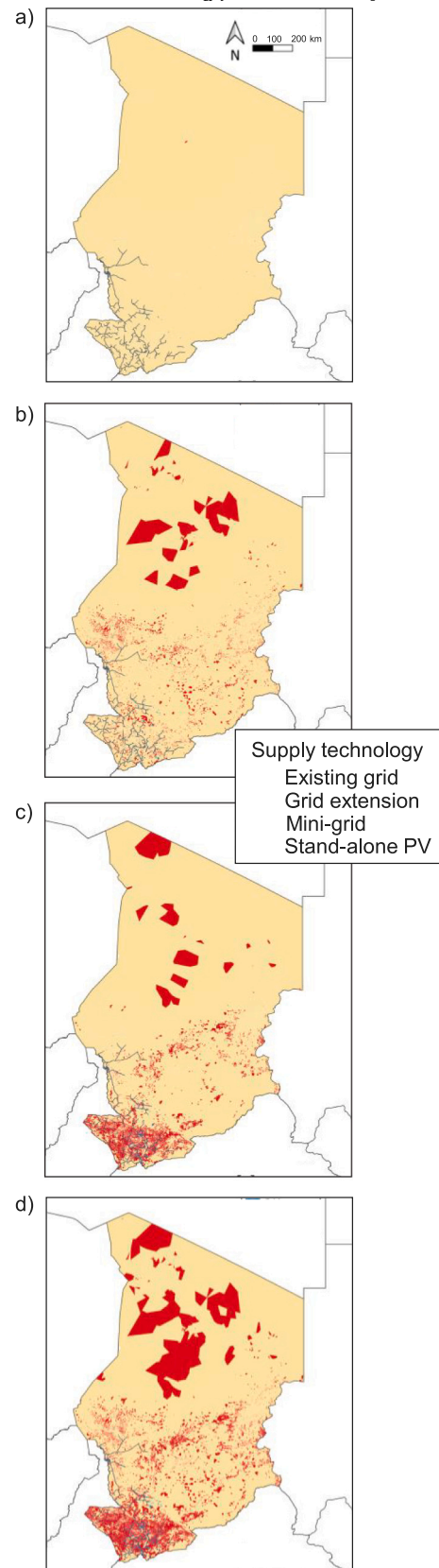


Fig. 7. Optimum electrification strategy for Chad until 2030: (a) Household demand only (b) Electric mobility & Household demand (c) Electric tractors & Household demand (d) all combined.

contrast, cropland favors installing MGs in remote areas. Especially the rural population gets attractive for higher capacity infrastructure

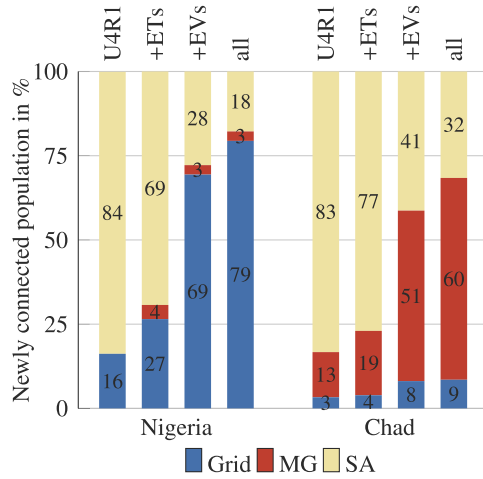


Fig. 8. Optimum electrification strategy dependent on the scenarios in Nigeria and Chad for an electrification rate of 100% until 2030.

in the case of electric tractors, as the additional demand comes from the cultivated areas and not from the settlements. This turns out to be utterly different in Chad, where almost no grid infrastructure exists and is barely economically feasible to be installed. Our results show the significance of MGs, which can electrify with least cost if EVs and ETs are implemented, despite the sparse population, the small fraction of cropland, and long distances to existing grid infrastructure.

Cost of electrification

Fig. 9 shows the relative percentage change in LCOE in Nigeria for the newly connected population: ETs reduce the cost per kilowatt-hour by a median of 49% and EVs by 54%. Combining both leads to a reduction of 54% with low spread. The results prove the hypothesis that the higher demand of the PUEs in the electricity infrastructure brings advantages to all consumers of the infrastructure by lower LCOE. The reduced cost of access to electricity is in favor of the overall goal of SDG 7.

The significant decrease in LCOE substantially lowers the electricity costs for households and businesses to run on lower electricity costs. Introducing the proposed PUEs reveals a stark contrast compared to the base case scenario, which considers only household demand (U4R1). The application of ETs shows a bigger spread, with the potential of lowering the LCOE up to 91%, because, in the case of settlements that do not have cropland, there is also no reduction in LCOE. As opposed to EVs, the demand of ET depends on the available cropland per settlement and does not correlate to the household demand, which is dependent on the population. The demand of EVs instead is rooted in the population density, similar to the household demand. Hence, the demand for EVs shows a lower spread than for ETs. Ultimately, the integration of both EVs and ETs aims to reduce LCOE universally, irrespective of cropland or population density, thereby benefiting all communities.

The observed trend highlights the benefits of mobile PUEs in lowering the energy costs for all and is consistent with the findings in literature (Booth et al., 2018; Wamalwa et al., 2023)

Sensitivity analysis

The model results presented show the maximum potential of the proposed PUEs' application. The currently limited data basis forces to estimate the large-scale demand for mobility and mechanization needs with the proposed methodology, along with their associated energy consumption. However, to enhance the reliability of our findings, account for any uncertainties in the data, as well as in the temporal

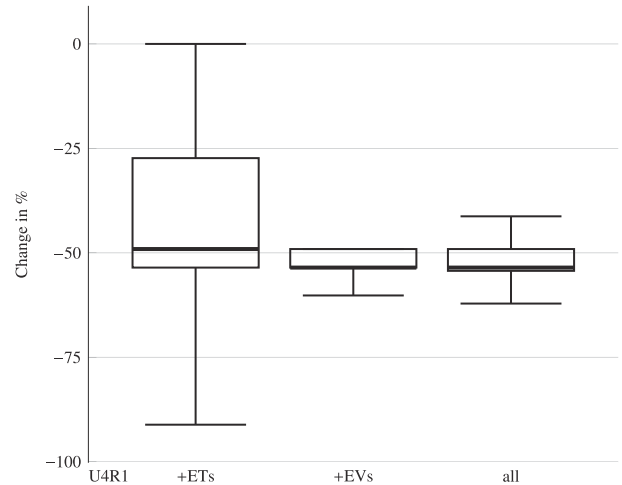


Fig. 9. LCOE in comparison to exclusively household demand being supplied (U4R1 as per the MTF) for each settlement in Nigeria.

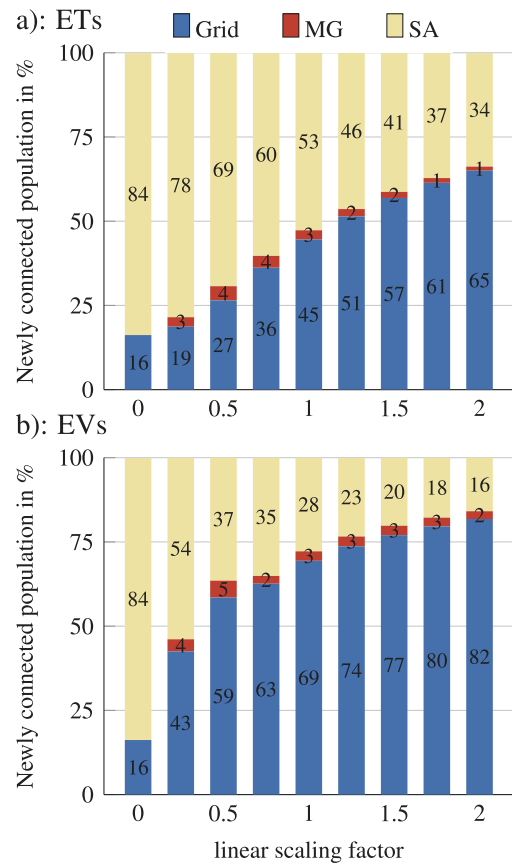


Fig. 10. Sensitivity of optimum electrification mode split in Nigeria with joint scaling of linear demand factors for (a) ETs and (b) EVs.

and spatial resolution of the estimated demands, and the underlying assumptions, we conduct a sensitivity analysis. This analysis involves linearly scaling the demand for EVs and ETs in Nigeria, demonstrating the impact of our demand estimation on the results and the flexibility in the methodology.

Electric tractor demand

As stated in Section "Spatial electric agriculture demand estimation", the demand intensity per given cropland depends on various factors such as the number and type of operations or the consumption of ETs per hectare. We estimated the demand in accordance with the state of the art with a constant rate of mechanized operations per

year as well as a constant electricity consumption per hectare. To show the influence on the results, we vary the demand by multiplying Eq. (3) by a multiple (Fig. 10(a)). The linear scaling factor 0 shows pure household demand (U4R1), while factor 1 corresponds to the case presented in Section “Results”. As the areal demand increases, the share of the population being electrified by grid extension increases also at the expense of SA systems. The linear scaling factor for the areal demand also has a roughly linear effect on the electrification strategy. The centralized network infrastructure can exploit economies of scale with higher demand. Hence, grid extension becomes more and more economically attractive for settlements further away from the existing grid, as a higher demand justifies greater investment for longer transmission lines. MGs can be seen as an intermediate technology for medium consumption levels and/or settlements further away from the grid. Asymptotically, grid extension covers all the electricity needs for agriculture due to its high demand; the remaining third of the population of Nigeria lives in settlements without cropland and is covered by SA systems.

Electric vehicle demand

As shown in Section “Spatial electric mobility demand estimation”, the potential electricity demand for EVs is calculated by Eq. (2) and relies on the mileage covered and constant consumption. Those factors reflect assumptions from literature about the specific consumption from a certain vehicle type, the load factor of each vehicle, and the total distance traveled per household, which all have a linear impact. Fig. 10(b) shows the dependence of the electrification strategy on the linear factors: As with ETs, the scaling factor 0 and 1 are the (U4R1) and unchanged PUE scenario. It becomes apparent that also with the increasing demand of EVs, the grid becomes more cost-effective compared to SA systems. MGs are the least-cost option for a nearly constant share and suit as a transitional solution. As EV demand correlates with population and, therefore, also with areas close to the existing grid, the share of new grid extension is generally higher than with ETs. Even though a lower demand (corresponding to scaling factor < 0.5) already significantly impacts the strategy, the gradient of change is flatter than for ETs. Asymptotically, the grid extends to cover most of the rural population, which is expected as the mobility demand concerns everyone living in rural settlements.

Discussion

This section sheds light on the reliability of input data, simplifications, assumptions made during the modeling process, and the resulting limitations to accuracy and transferability for the presented results both generally and for the results presented. The section underlines the theoretical nature of the study and shows the limitations of the top-down perspective on the analyzed effects of the technology leap.

Order of magnitude

We aim to investigate the maximum theoretical impact on the optimum electrification strategy the considered PUEs can have and give an estimate for the order of magnitude. We are constrained to estimate the maximum demand based on geographical factors, which is, in the case of EV, the mobility demand per household and, in the case of ET, the arable land per settlement. The demand for mobility and mechanization might vary in reality and will not necessarily be purely satisfied by EVs and ETs. To highlight the maximum potential impact that joint consideration of these PUEs can have on national electrification strategies, we assume complete electrification of both mobility and agriculture PUEs. This study neither considers the transitional process of adopting the PUEs, nor the extra investment to unlock the joint consideration. The practical implementation of our approach requires a detailed local analysis per settlement, considering updated demand scenarios, socio-economic factors, operational strategies, and cost of capital. To account for the uncertainty in our demand estimation, Section “Sensitivity analysis” expresses the sensitivity.

Necessary Investment

The implementation of higher capacity electrification modes as well as the adoption of EVs and ETs requires significantly higher upfront investment. Even if the concept of LCOE takes into account the amortization of higher upfront investments, economies with high capital costs do not benefit from the higher return on higher upfront investments. We want to remark that advent of ETs and EVs offers the possibility to leapfrog the fossil-fuel dependency in agriculture and transportation, however, the success and deployment in rural areas requires the co-planning of electrification infrastructure and mechanization programs. As remarked by Wamalwa et al. (2023), the success of such a connection requires “strategic cross-sectoral partnerships of all the involved actors-including relevant government institutions, technology providers, MGs developers, and financial institutions”. Hence, this study aims to show policymakers the strategic potential in EVs and ETs to ease the way to mechanization, rural mobility, and access to electricity.

Electricity demand profile

We account for fluctuating demand of household, EVs and ETs through the available base-to-peak factor of OnSSET. We are constrained to neglect the seasonality of demand, which influences the dimensioning and profitability of energy systems, since OnSSET calculates with the annual electricity demand. ETs but also EVs will experience seasonal as well as daily variations in load. The daily variations depend on the charging technology and can be controlled differently with either battery swapping or direct conductive charging. For a more detailed analysis of the seasonality, the availability of ETs and especially the differing energy consumption for the different mechanized operations must be known. This is suitable for in-depth analysis of individual settlements that require more modeling, leading to results with lower planning errors and more accurate cost estimates, but does not allow for a scalable method like ours.

Demand estimation

As there are no spatially resolved models for mobility and mechanization requirements in SSA, we propose a method based solely on available geographic data. Mobility data, especially from rural areas, is the key to solving such chicken and egg problems, adaptively designing infrastructure on a local level, and validating spatial modeling approaches at the national level. We agree to Collett and Hirmer (2021) that data is needed to decarbonize mobility modes in SSA to be able to refine our geospatial estimation method.

The same counts for the geospatial data used to estimate agricultural demand: Since the data on the area under cultivation does not yet differentiate by type of crop, the actual mechanization potential and the resulting energy demand must be estimated on the basis of the size of the area under cultivation. Similarly, the assignment of cropland to settlements through Voronoi cells emphasizes surrounding settlement density as the key factor in allocation and might neglect settlement characteristics, such as the number of actual farmers. Partition using population-weighted Voronoi cells could shift the focus onto population density. While this reflects the settlements’ actual cultivation capability better, it still does not consider any value chain limitations. Furthermore, we assign charging loads to the settlement where the corresponding mobility or mechanization demand is located. If regional operational bases might be established, from which both EVs and ETs could operate, this is not necessarily the case. Hence, we propose for a local implementation that such patterns should be evaluated in case studies and field trials.

Contextual influence

The existing narrative that Africa can avoid a phase of fossil fuel dependency in electricity generation (Peters et al., 2024) needs to account for the implementation of electric mobility and mechanization and their impact on the renewable energy pathways, for which this publication offers a theoretical investigation. However, actual implementation demands detailed efforts at the local level, accounting for a wide range of contextual parameters, including technical aspects (e.g., suitability of

commercially available vehicles), economic considerations (e.g., access to capital), and socio-economic factors (e.g., reliance on informal transport modes, willingness and ability to pay for electricity). Therefore, any insights drawn from this study should be carefully contextualized by considering additional local evidence and integrating local scholars.

Future work

To advance our work, future research will investigate further impact criteria except in addition to the LCOE, including total investment for the electrification strategies and the cost of the implementation of ETs. Another potential study area is the detailed interaction of the proposed PUEs with the different electrification systems and what further synergies can be unlocked in joint design and operation, for instance, through battery swapping and rental systems. This might include pay-as-you-go business models, accounting for customers' willingness to pay. For electric tractors, in particular, which are still at an early stage of development, the usage patterns and the resulting optimum design of the power train for various operational conditions are required to be able to show more exact values for energy consumption.

Conclusion

The economically sustainable and widespread electrification of rural areas in Sub-Saharan Africa (SSA) requires stimulation of demand through Productive Uses of Electricity (PUEs) and a coordinated strategy on where to deploy which form of electrification (grid extension, decentralized mini-grids (MGs) or standalone systems (SA)). With the advent of electric tractors (ET) and electric vehicles (EV) in SSA, our study is the first to consider such a mobile PUE. The aim of the study is a theoretical investigation of the impact and potential between our proposed PUE and the electrification strategy.

The results of our study show that the widespread application of EVs and ETs and their energy consumption does require the investment into higher capacity electrification modes (grid extension and MG). However, we show in the example of Nigeria that by applying EVs, the leveled cost of electricity (LCOE) is reduced on average by 54%, and by the implementation of ETs a reduction of 49% is reached. The demand of EVs favors areas with high population by changing the electrification strategy for around 56% of the unelectrified population. The demand of ETs is influenced by the available cropland per settlement and changes the electrification strategy for 15% of the unelectrified population, but has an impact on 56% of the total cropland area. Since agriculture is of particularly crucial importance for the region's economic growth and still holds great potential for increasing productivity, the results show that access to electricity should be increased in economically interesting areas while at the same time increasing productivity with ETs.

For this investigation, we developed a methodology for scalable electricity demand estimation for both EVs and ETs per settlement in the target region of SSA, which advances the Open Source Spatial

Electrification Tool (OnSSET). The results are based on spatial information from the Copernicus Land Cover data and population data. To cope with limited data availability, our method determines the mobility demand of individual households by electric motorcycles by a mix of constant typical local destinations and the distance to the nearest paved road as a spatially variable component covered. The tractor demand is based on the available cropland and area-related energy consumption of ETs. With the prevalence of smallholder farming, we apply Voronoi partitioning to spatially allocate the cropland to the nearest settlement for each field.

In conclusion, for a technology leapfrogging of fossil fuel-driven vehicles in agriculture and transport in SSA the large impact of ETs and EVs underlines the necessity to integrate them and their rollout timeline into electrification planning and strategy for every nation to maximize the effect of investments and the usability of supply. Additionally, a multitude of locally refined case studies is needed who publish their insights publicly.

This work can be suited as a recommendation for policy makers to create the environment to leverage the analyzed potential and therefore account for the changes needed in electricity supply option for EVs and ETs dependent on the desired mobility and mechanization strategies.

CRedit authorship contribution statement

Korbinian Götz: First author, Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Philipp Rosner:** Writing – review & editing, Writing – original draft, Validation, Methodology. **Irénée Froissart:** Writing – review & editing, Software, Methodology. **Markus Lienkamp:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

The parametrization for demographic, technical and economic costs can be found in [Tables 3–5](#).

Table 3

Parametrization of OnSSET in accordance to [World Bank Group \(2019\)](#).

Parameter	Value	Source
Start year	2020	
End year	2030	
Population 2020	208 330 000	World Bank Group (2019)
Urban population ratio 2020	0.52	
Expected Population 2030	262 258 000	World Bank Group (2019)
Urban population ratio 2030	0.52	
Number of people per Household urban	4	Mentis et al. (2017) and World Bank Group (2019)
Number of people per Household rural	4.6	Mentis et al. (2017) and World Bank Group (2019)
Expected investment cost per kW of additional grid capacity	2238	
Expected losses in the grid network	0.167	Agutu et al. (2022)
Average to peak load ratio for the grid; used for sizing additional capacity due grid extension and to accommodate reliability issues	0.8	
Percentage of capital cost increase in each grid extension iteration; used to accommodate reinforcement of grid and reliability of supply	0.1	

(continued on next page)

Table 3 (continued).

Parameter	Value	Source
Maximum distance that MV lines can reach in each iteration loop	50 km	
Capacity that can be added to the central grid per year of analysis	99	
Official national electrification rate in the base year	0.57	World Bank Open Data (2024a)
Official national electrification rate in rural areas in the base year	0.31	World Bank Open Data (2024a)
Official national electrification rate in urban areas in the base year	0.82	World Bank Open Data (2024a)
Modeled national electrification rate in the base year	0.57	
Modeled national electrification rate in rural areas in the base year	0.31	
Modeled national electrification rate in urban areas in the base year	0.82	
Diesel Cost	1 USD/l	
Grid cost low scenario	0.052	Taliotis et al. (2016)
Grid cost high scenario	0.065	Taliotis et al. (2016)
Grid connections limit	5 208 000 (=2.5% * PopStartYear)	World Bank Group (2019)

Table 4
Parameters for off-grid technology investment costs (Carlin et al., 2018; Johannsen et al., 2020; Korkovelos et al., 2019, 2018; World Bank, 2024b; World Bank Group, 2019).

Supply option	Parameter	Value
Hydro MG	Global investment cost	3000 USD/kWp
Hybrid MG	PV panels	503 USD/kWp
	Batteries	139 USD/kWh
	Inverter	80 USD/kWp
	Charge controller	142 USD/kW
	Diesel generator	261 USD/kW
	Wind turbine	2800 USD/kW
SA	If $P_{I,off-grid} < 0.02$ kW	9620 USD/kWp
	$0.02 < P_{I,off-grid} < 0.05$ kW	8780 USD/kWp
	$0.05 < P_{I,off-grid} < 0.1$ kW	6380 USD/kWp
	$0.1 < P_{I,off-grid} < 1$ kW	4470 USD/kWp
	$P_{I,off-grid} > 1$ kW	6950 USD/kWp

Table 5
Grid generating and transmission & distribution cost.

Category	Parameter	Value	Source
Grid	On-grid cost low	0.052 USD/kWh	Chattopadhyay (2018), Pappis (2016), Taliotis et al. (2016) and World Bank Group (2019)
	On-grid cost high	0.065 USD/kWh	
Transmission & distribution	HV line (69–132 kV)	53 000 USD/km	
	MV line (11–33 kV)	7000 USD/km	Aissa et al. (2024), International Energy Agency (2017), Korkovelos et al. (2017, 2019), Mentis et al. (2017), Nerini et al. (2016), van Ruijven et al. (2012), World Bank (2024a, 2024c) and World Bank Group (2019)
	LV line (0.2–0.4 kV)	4250 USD/km	
	HV to MV substation	25 000 USD/unit	
	MV to LV substation	10 000 USD/unit	
	Service transformer	4250 USD/unit	

Appendix B Supplementary data

Supplementary data on the results is provided here: <https://doi.org/10.14459/2024mp1749237>.

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